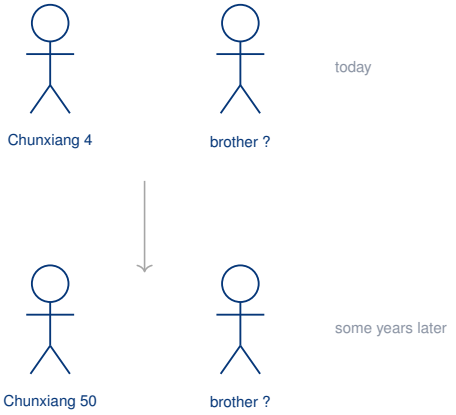
XIAMEN UNIVERSITY MALAYSIA MATHEMATICS STUDENT COUNCIL

No. 01

Age Difference Puzzle

Theme 1: Finance / Game / Logic

Chunxiang is 4 years old now. His younger brother is half his age. When Chunxiang is 50 years old, how old will his younger brother be?



Record your final numerical answer carefully.

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No. 02

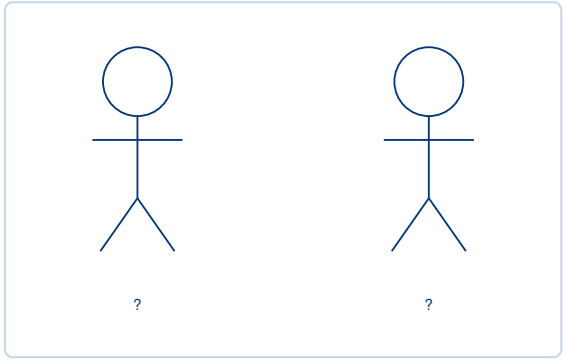
Two-Child Probability Puzzle

Theme 1: Finance / Game / Logic

A family has two children. Given that at least one of them is a boy, what is the probability that the other child is also a boy?

Answer as a fraction.

two children, at least one of them a boy



Think about all equally likely possibilities.

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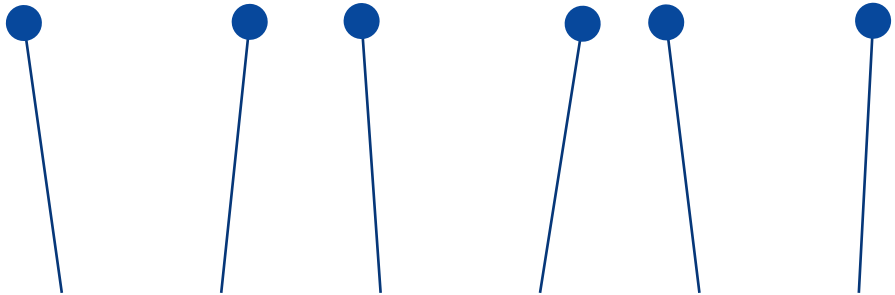
No. 03

Six-Matchstick Triangle Challenge

Theme 1: Finance / Game / Logic

Using exactly 6 matchsticks, what is the maximum number of equilateral triangles you can form?

You may use physical matchsticks to test your idea.



six matchsticks, all the same length

Hint: Do not assume the construction must be flat.

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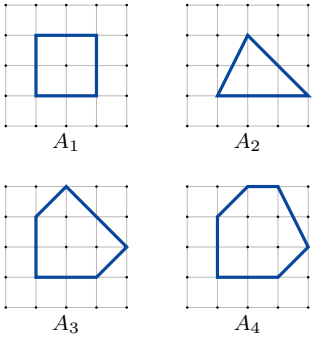
Scan to open the answer submission form

Pick's Theorem Area Challenge

Theme 4: Engineering / Sciences / Math

The four polygons below have vertices on lattice points. Find

$$A_1 + A_2 + A_3 + A_4.$$



Count the lattice points on each boundary and the ones strictly inside. That is all you need.

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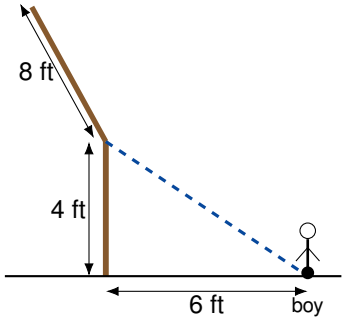
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Will the Boy Be Safe?

Theme 4: Engineering / Sciences / Math

A tree breaks 4 ft above the ground. The broken upper part is 8 ft long. A boy is standing 6 ft from the base of the tree.
If the falling part forms a right triangle with the ground, will the boy be safe?

Answer with how far from the base the treetop reaches, in feet, to one decimal place.



Use only elementary geometry; no complicated physics is required.

SCAN TO SUBMIT

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XIAMEN UNIVERSITY MALAYSIA MATHEMATICS STUDENT COUNCIL

No. 06

Infinite Radical Challenge

Theme 4: Engineering / Sciences / Math

Find the value of each expression. Give **(a)** as a fraction, **(b)** to three decimal places and **(c)** as a whole number.

(a) $\sqrt{1.77777\ldots}$

(b) $\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\cdots}}}}$

(c) $\sqrt{2\sqrt{4\sqrt{8\sqrt{16\sqrt{\cdots}}}}}$



each one holds a smaller copy of itself

Look for a self-similar structure.

SCAN TO SUBMIT

[QR CODE PLACEHOLDER]

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No. 07

Mirror Symmetry Challenge

Theme 3: Arts / Literature / TCM

Hold each of these words up to a mirror. Some come back word for word. The rest come back as nonsense.

MATH	WOW	NOON	TOOT
LEVEL	AHA	RADAR	MUM

Which of the eight read the same in the mirror?



A mirror does two things at once: it flips every letter, and it reverses their order.

SCAN TO SUBMIT

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No. 08

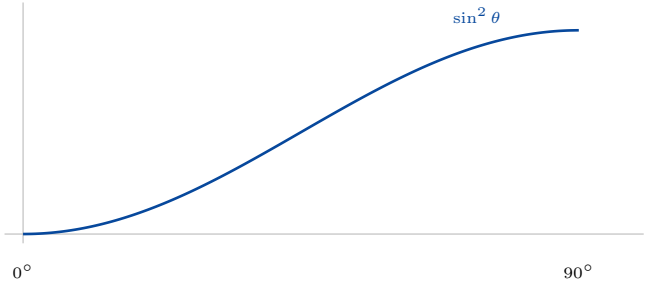
Patterned Sum Challenge

Theme 4: Engineering / Sciences / Math

Evaluate:

$$\sin^2 0^\circ + \sin^2 1^\circ + \cdots + \sin^2 89^\circ + \sin^2 90^\circ$$

$$100 - 98 + 96 - 94 + 92 - 90 + \cdots + 8 - 6 + 4 - 2$$



Try pairing terms instead of calculating one by one.

SCAN TO SUBMIT

[QR CODE PLACEHOLDER]

Scan to open the answer submission form

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No. 09

Find the Fallacy

Theme 4: Engineering / Sciences / Math

Each "proof" below reaches an impossible conclusion. In each one, find the first line that is false.

Answer with the three line numbers in order, as one string, for example 123**.**

A. Why $\pi = 3$?

1. $x = \frac{\pi+3}{2}$
2. $2x = \pi + 3$
3. $2x(\pi - 3) = (\pi + 3)(\pi - 3)$
4. $2\pi x - 6x = \pi^2 - 9$
5. $9 - 6x + x^2 = \pi^2 - 2\pi x + x^2$
6. $(3 - x)^2 = (\pi - x)^2$
7. $3 - x = \pi - x$
8. $\pi = 3$

B. Why $2 = 1$?

1. $a = b$
2. $a^2 = ab$
3. $a^2 + a^2 = a^2 + ab$
4. $2a^2 - 2ab = a^2 - ab$
5. $2(a^2 - ab) = 1(a^2 - ab)$
6. $2 = 1$

C. Why $4 = 5$?

1. $4 = 4 - \frac{9}{2} + \frac{9}{2}$
2. $= \sqrt{(4 - \frac{9}{2})^2} + \frac{9}{2}$
3. $= \sqrt{(5 - \frac{9}{2})^2} + \frac{9}{2}$
4. $= 5 - \frac{9}{2} + \frac{9}{2}$
5. $= 5$

SCAN TO SUBMIT

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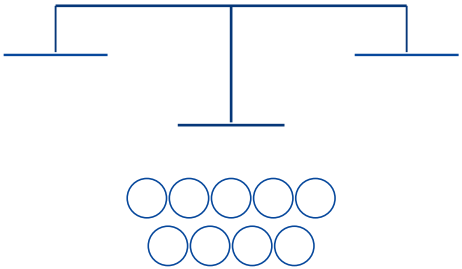
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The Fake Coin

Theme 1: Finance / Game / Logic

There are 9 identical-looking coins, and exactly 1 of them is fake. The fake coin has a different weight, but you do not know whether it is heavier or lighter than a real coin.

Using a balance scale, what is the minimum number of weighings needed to guarantee finding the fake coin **and** determining whether it is heavier or lighter?



9 coins, one of them fake

Your strategy must work in every possible case.

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Palindrome Pattern

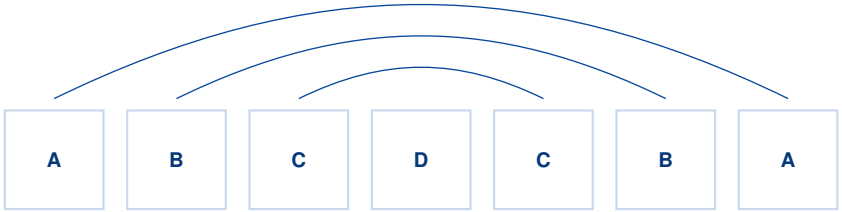
Theme 3: Arts / Literature / TCM

A 7-letter word follows the symmetrical pattern

$$A\,B\,C\,D\,C\,B\,A.$$

The 1st and 7th letters are the same, the 2nd and 6th are the same, and the 3rd and 5th are the same.

If you may use any of the 26 standard English letters, how many different 7-letter combinations can be formed using this pattern?



Count how many positions can be chosen independently.

SCAN TO SUBMIT

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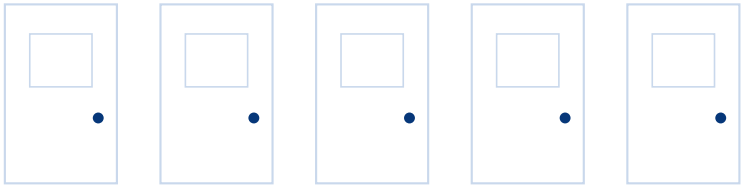
Passcode Collision

Theme 2: AI / Data Science

A housing estate has 300 apartments. Each door uses a 4-digit passcode from 0000 to 9999, so there are 10,000 possible passcodes.

Assuming each apartment chooses its passcode independently and uniformly at random, what is the probability that **at least two apartments** share exactly the same passcode?

Answer as a percentage, to the nearest whole number.



same code, two doors

Think of the birthday problem in a different setting.

SCAN TO SUBMIT

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No. 13

Mean vs. Median

Theme 2: AI / Data Science

Nine local working-class individuals are sitting in a café in Malaysia. Each earns RM 60,000 per year.
Suddenly, a top C-Suite executive earning RM 12,000,000 per year joins them.

For the ten of them together:

1. What is the new mean annual salary, in RM?
2. What is the new median annual salary, in RM?



Which measure is more affected by an extreme value?

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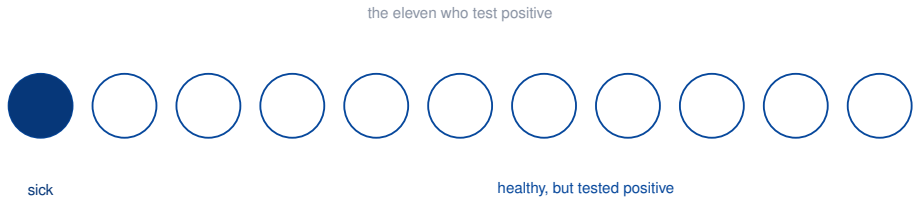
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Rare Disease and False Positives

Theme 3: Arts / Literature / TCM

In a city of 1,000 people, only 1 person has a rare disease. A hospital tests everyone.
The sick person tests positive. The test also incorrectly labels about 1% of healthy people as positive, so about 10 healthy people also test positive.
Therefore, about 11 people receive a positive result, but only 1 is actually sick.

A patient receives a positive result. **What is the probability that they are actually sick? Answer as a percentage, to the nearest whole number.**



So what should the doctor do? Test again: a second independent positive pushes the same calculation the other way.

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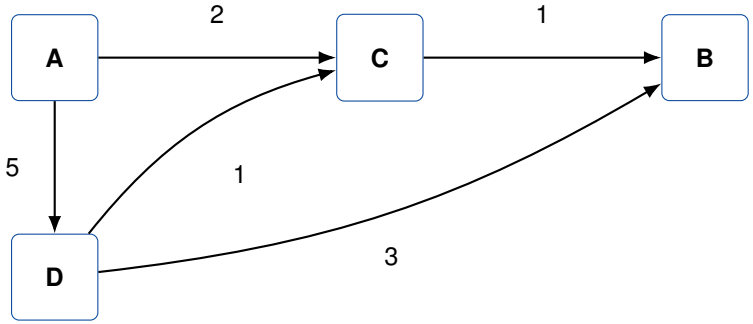
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Shortest Path for a Delivery Robot

Theme 2: AI / Data Science

A delivery robot must travel from A to B. Each directed road has a travel time in minutes.



What is the shortest possible travel time, in minutes?

This illustrates the idea behind Dijkstra's shortest-path algorithm.

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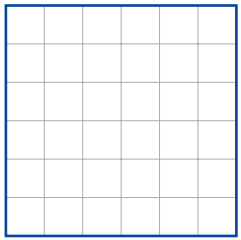
Mondrian Art Problem

Theme 3: Arts / Literature / TCM

Fill the 6×6 grid completely using non-overlapping rectangles. Squares are allowed.

- You must use at least two rectangles.
- No two rectangles may have the same dimensions.
- A 2×3 rectangle and a 3×2 rectangle count as the same dimensions.

Your score is the area of the largest rectangle minus the area of the smallest. **What is the lowest score that is possible?**



Use the grid to try it out, then say how low the score can go.

SCAN TO SUBMIT

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No. 17

Equation to 45

Theme 2: AI / Data Science

Fill the rows with true equations worth exactly $\boxed{45}$, one symbol per cell. Then count how many there are of this shape:

$$\square\square \circ \square = 45$$

a two-digit number, one operation, and a single digit (0 to 9, and 0 counts).

How many such equations are there altogether? Not just the ones you wrote: every one that exists. Count $+$, $-$, $\times$ and $\div$ separately.

The $=$ sign takes a cell of its own. Work the four operations out one at a time.

SCAN TO SUBMIT

[QR CODE PLACEHOLDER]

Scan to open the answer submission form

Target 24

Theme 2: AI / Data Science

Make exactly $\boxed{24}$ from

$$\boxed{1} \quad \boxed{10} \quad \boxed{12} \quad \boxed{6}$$

- Use **every** number, and each one **exactly once**.
- Combine two numbers at a time with $+$, $-$, $\times$ or $\div$. The result replaces them both, so four numbers become three, then two, then one, and that last number must be 24.
- Use an operation as often as you like, or not at all. No other numbers may be brought in.



four numbers become three, then two, then one

Write down the order you combined them in. That is your answer.

SCAN TO SUBMIT

[QR CODE PLACEHOLDER]

Scan to open the answer submission form

Cryptogram Puzzle

Theme 3: Arts / Literature / TCM

A message has been encoded using a monoalphabetic substitution cipher: every occurrence of the same plaintext letter is replaced by the same ciphertext letter.

Decode the hidden message:

```
QLSJFQLSKAZ KZ SJF WOFFT UG SJF ZAKFTAFZ
```

You may use letter-frequency patterns, repeated letters and word structure.

how often each letter appears



Hint: F is the most common letter here, and E is the most common letter in English.

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The Monty Hall Problem

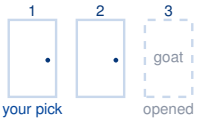
Theme 1: Finance / Game / Logic

You are a contestant on a game show with 3 closed doors.
Behind one door is a sports car; behind the other two are goats.

1. You randomly choose Door 1.
2. The host, who knows what is behind every door, opens Door 3 and reveals a goat.
3. The host asks whether you want to stay with Door 1 or switch to Door 2.

Does switching increase your chance of winning the car?

Answer with the probability of winning if you switch, as a fraction.



Almost everyone gets this one wrong the first time.

SCAN TO SUBMIT

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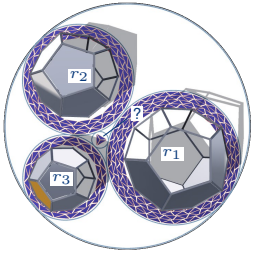
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Kissing Circles

Theme 4: Engineering / Sciences / Math

Installation 2 stands on the round glass floor beside you. Each of the three mirror dodecahedra sits inside its own circle inked on the floor, and the three circles touch one another and the edge of the floor: $r_1 = 178.2$ cm, $r_2 = 148.8$ cm and $r_3 = 112.9$ cm.

A fourth, small circle is inked in the gap in the middle, touching all three. **What is its radius, to the nearest centimetre?** Pace it out afterwards and see.



Hint (Descartes' Circle Theorem): for four mutually tangent circles with $k_i = 1/r_i$, $(k_1+k_2+k_3+k_4)^2 = 2(k_1^2+k_2^2+k_3^2+k_4^2)$. The floor's own edge solves it too, counting its curvature as negative.

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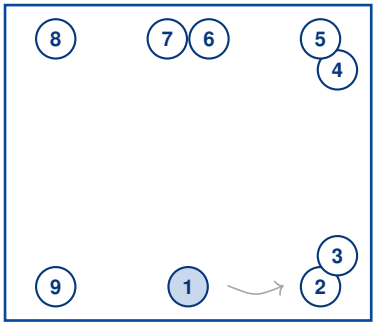
The Josephus Problem

Theme 1: Finance / Game / Logic

Nine people stand around the edge of the room, numbered 1 to 9 clockwise. Counting clockwise from 1, every second person leaves: 1 stays, 2 leaves, 3 stays, 4 leaves, and so on.

The count **never restarts**. It just carries on round the circle over whoever is still in it (so after 8 leaves, 9 stays and 1 leaves) until one person is left.

Where do you stand to be that person?



Try it with 4, 5 and 6 people first, then look at what happens whenever the head count is a power of 2.

SCAN TO SUBMIT

[QR CODE PLACEHOLDER]

Scan to open the answer submission form

Mystery Chest Decode Card

Final Challenge

Each card below has one whole-number answer. Copy your answers in, then work out the four digits.

N_1	No. 04	Pick's Theorem: the total area	
N_2	No. 06	Infinite Radical: part (c)	
N_3	No. 08	Patterned Sum: the second expression	
N_4	No. 11	Palindrome Pattern	
N_5	No. 12	Passcode Collision: as a whole %	
N_6	No. 13	Mean vs. Median: the new mean (RM)	
N_7	No. 15	Delivery Robot: minimum minutes	

$$D_1 = (N_1 + N_2) \bmod 10, \quad D_3 = (N_5 + N_6) \bmod 10,$$
$$D_2 = (N_3 + N_4) \bmod 10, \quad D_4 = N_7 \bmod 10.$$

Password $D_1D_2D_3D_4$: ☐ ☐ ☐ ☐

$x \bmod 10$ is the last digit of x, so only the last digit of each answer matters.

MYSTERY CHEST

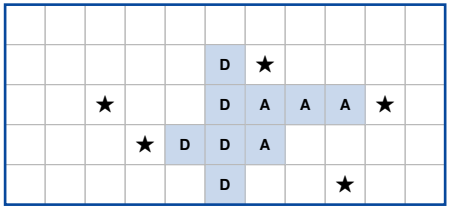
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Use your recorded answers to work out the password

Kanoodle: Floor Edition

Theme 1: Finance / Game / Logic

Pieces **A** and **D** are already placed. Fill the whole 5×11 board with all 12 pieces: no overlaps, no gaps and nothing left over.



Now read the letters of the five pieces covering the ★ cells, in reading order (top row first, left to right). Those five letters are your passcode.

There is exactly one way to finish this board, so everyone who solves it reads the same five letters.

ENTER YOUR PASSCODE

[QR CODE PLACEHOLDER]

Scan to open the Math Week site, sign in and enter your passcode

Tower of Hanoi

Theme 2: AI / Data Science

Four disks start stacked on peg **A**; disk 1 is the smallest. Pegs are **A**, **B**, **C** from left to right.

- Move one disk at a time.
- A disk may never be placed on a smaller disk.

Move the whole tower from peg **A** to peg **C** in the fewest possible moves, counting as you go.

In that shortest solution: **which disk moves on the 5th move, and onto which peg?**

Your passcode is the disk number followed by the peg letter (format: `4B`).



The shortest solution takes $2^4 - 1 = 15$ moves and there is only one of them, so every move in it is forced.

ENTER YOUR PASSCODE

[QR CODE PLACEHOLDER]

Scan to open the Math Week site, sign in and enter your passcode

Bagua: The Round Table of Eight

Theme 3: Arts / Literature / TCM

Each trigram is a 3-bit binary number read **bottom line first**: a solid line is **1**, a broken line is **0**. The small number under each one is its Later Heaven number.



Seat all eight trigrams around the eight marked seats, numbered 1 to 8 clockwise, so that all four clues hold:

1. Between every seat and the next one clockwise, seat 8 back to seat 1 included, **exactly one line changes**.
2. 111 sits in seat 1.
3. The trigram in seat 5 has exactly one solid line, and it is the *middle* line.
4. From seat 1 to seat 2, the line that changes is the *bottom* line.

Passcode: the Later Heaven numbers of the three trigrams that have exactly one solid line, read clockwise from seat 1.

Check yourself: the Later Heaven numbers in seats 1, 3, 5 and 7 must add up to 18.

ENTER YOUR PASSCODE

[QR CODE PLACEHOLDER]

Scan to open the Math Week site, sign in and enter your passcode

Liar Dialogue

Theme 4: Engineering / Sciences / Math

Nailong always tells the truth. **LY3** always lies. They are talking about a hidden **two-digit** number.

Nailong: "The two digits add up to ten."

LY3: "The number is even."

Nailong: "The number is greater than sixty."

LY3: "The number is prime."

What is the hidden number? Your passcode is that number.



Every statement tells you something. A lie just tells you the opposite.

ENTER YOUR PASSCODE

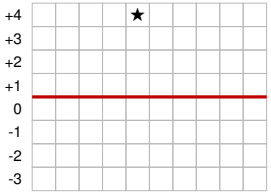
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Scan to open the Math Week site, sign in and enter your passcode

Conway's Soldiers

Theme 1: Finance / Game / Logic

Soldiers start on cells **below** the line. One move only: a soldier jumps over an adjacent soldier, horizontally or vertically, onto the empty cell beyond, and the soldier jumped over is removed.



Place as many soldiers as you like below the line, then get a single soldier up to the **4th row above it**, the ★ row.

Answer with the smallest number of soldiers that can do it. Then try for the 5th row.

Bonus points if you follow `@xmummath` on Instagram.

Row 4 can be done. Row 5 cannot be done with any number of soldiers at all, and that can be proved.

ASK THE COMMITTEE

Nothing to scan here. A committee member watches your attempt and shows you a claim QR when you win.

The Knight's Tour

Theme 2: AI / Data Science

Starting from any square you like, move the knight so that it visits **every square of the board exactly once**.

Pick your board: the full 8×8, a taped-off 5×5 or 6×6 corner, or a board with a few squares marked as holes the knight may never land on.
A checker goes down on every square the knight leaves, so the path is there to see and a revisit shows up at once.

Answer with how many moves a full tour of the 8×8 board takes.



two along, one across

Bonus points if you follow `@xmummath` on Instagram.

Always move to the square with the fewest onward moves.

ASK THE COMMITTEE

Nothing to scan here. A committee member watches your attempt and shows you a claim QR when you win.

A Blind, A Deaf, A Mute

Theme 3: Arts / Literature / TCM

Three players, three restrictions:

Blind	wears an eye mask throughout, and gives the final answer.
Deaf	wears earmuffs and cannot hear anything.
Mute	may not speak at all.

The information needed is split up. The Mute and the Deaf each hold a hint card; the committee tells the Blind the remaining piece. **Nobody has all of it.**

Work together inside your restrictions. The Blind must state the answer within the time limit.



Blind



Deaf



Mute

Bonus points if you follow `@xmummath` on Instagram.

The hard part is not the mathematics. It is getting a number across.

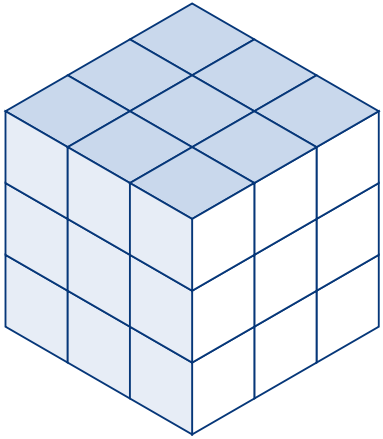
ASK THE COMMITTEE

Nothing to scan here. A committee member watches your attempt and shows you a claim QR when you win.

Rubik Mini Booth

Theme 4: Engineering / Sciences / Math

Complete the challenge.



Bonus points if you follow `@xmummath` on Instagram.

ASK THE COMMITTEE

Nothing to scan here. A committee member watches your attempt and shows you a claim QR when you win.

Luban Lock & Rope Puzzle Challenge

Theme 4: Engineering / Sciences / Math

A timed challenge in two tiers. The committee member keeps a solved reference copy of each, and starts the stopwatch when you do.

- **Lock tier:** the six-way Luban lock, handed to you in pieces. Put it back together inside the time limit, then **answer with how many pieces it has**.
- **Rope tier:** a string-and-ring puzzle. Free the ring and thread it back inside the time limit, then **answer with whether the cord has to be untied to do it**.

An assembly photo lies face down on the table. You may turn it over at any time; finishing without it is worth more.

Bonus points if you follow `@xmummath` on Instagram.

The lock is interlocking assembly: one order of insertion works and the last piece in is the key. The rope is topology: the ring moves by a continuous deformation of the cord, never by force.

ASK THE COMMITTEE

Nothing to scan here. A committee member watches your attempt and shows you a claim QR when you win.